

Usnea VCF filtering

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VCF QC viz for Usnea resequencing data

35 *Usnea florida* and *Usnea subfloridana* resequencing samples were aligned to the *U. florida* genome using BWA-MEM2, and SNPs were called using Freebayes. Here, 59,539 SNPs were randomly subsampled out of 4,961,854 SNPs for QC purposes. VCFtools was used to calculate several QC metrics visualized with ggplot below.

Set up

Load tidyverse and set working directory.

```
library(tidyverse)
```

```
## — Attaching core tidyverse packages — tidyverse 2.0.0 —
## ✓ dplyr      1.2.1      ✓ readr      2.2.0
## ✓ forcats    1.0.1      ✓ stringr    1.5.1
## ✓ ggplot2    4.0.3      ✓ tibble     3.3.1
## ✓ lubridate  1.9.5      ✓ tidyr      1.3.2
## ✓ purrr      1.0.2
## — Conflicts — tidyverse_conflicts() —
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
setwd("/home/FM/sblanco/GWAS_8072026/uflo/04_freebayes/QC")
```

Variant Quality

Phred quality score for each site. 26896 have QUAL < 30 (~50%).

```
var_qual <- read_delim("uflo_vcf_subset.lqual", delim = "\t",
                      col_names = c("chr", "pos", "qual"), skip = 1)
```

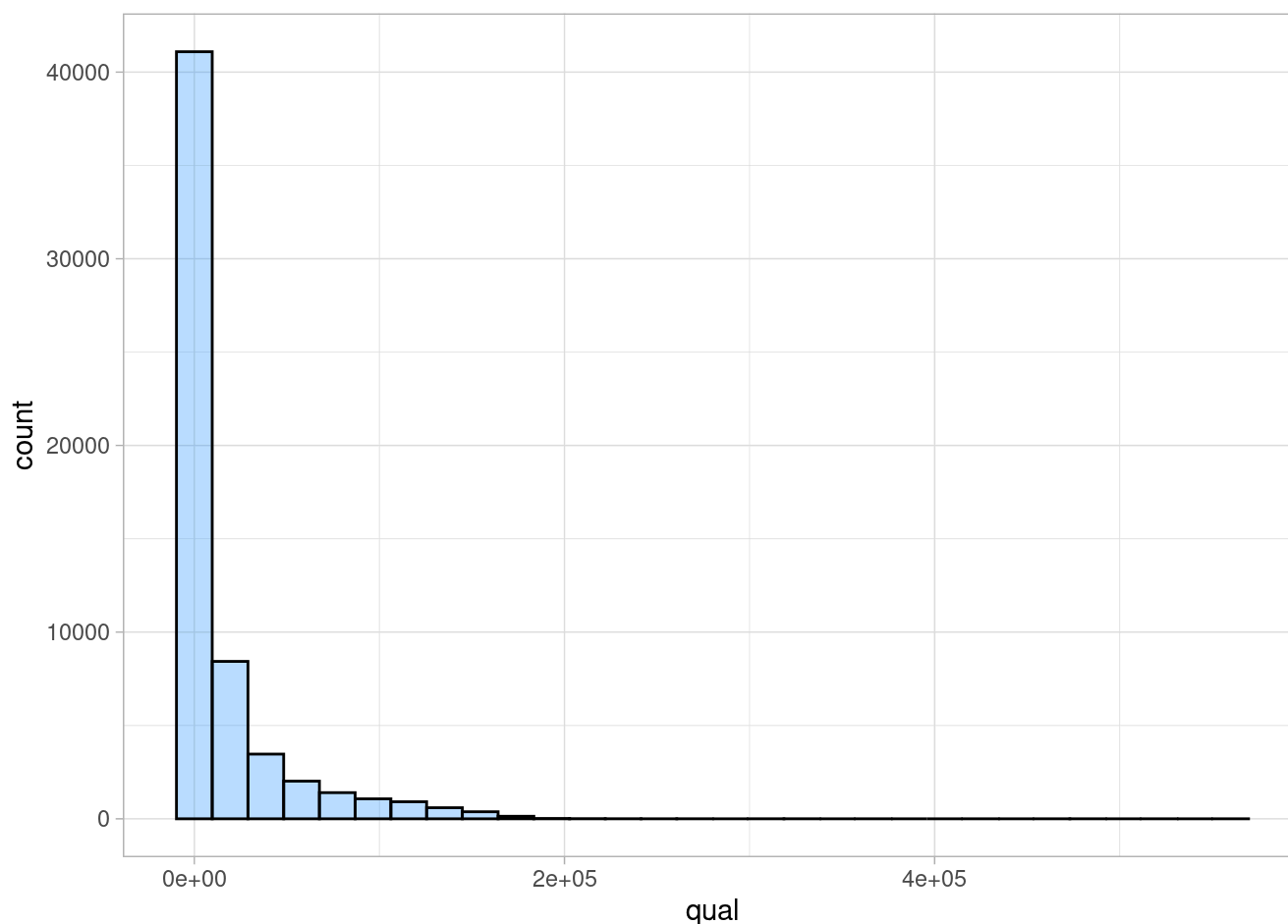
```
## Rows: 59539 Columns: 3
## — Column specification —————
## Delimiter: "\t"
## chr (1): chr
## dbl (2): pos, qual
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
summary(var_qual$qual)
```

```
##      Min.   1st Qu.   Median     Mean   3rd Qu.     Max.
##      0.0      0.0      650.8  15602.8  14969.1 559966.0
```

```
a <- ggplot(var_qual, aes(qual)) +
  geom_histogram(fill = "dodgerblue1", colour = "black", alpha = 0.3)
a + theme_light()
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



Decision: set min QUAL to 30.

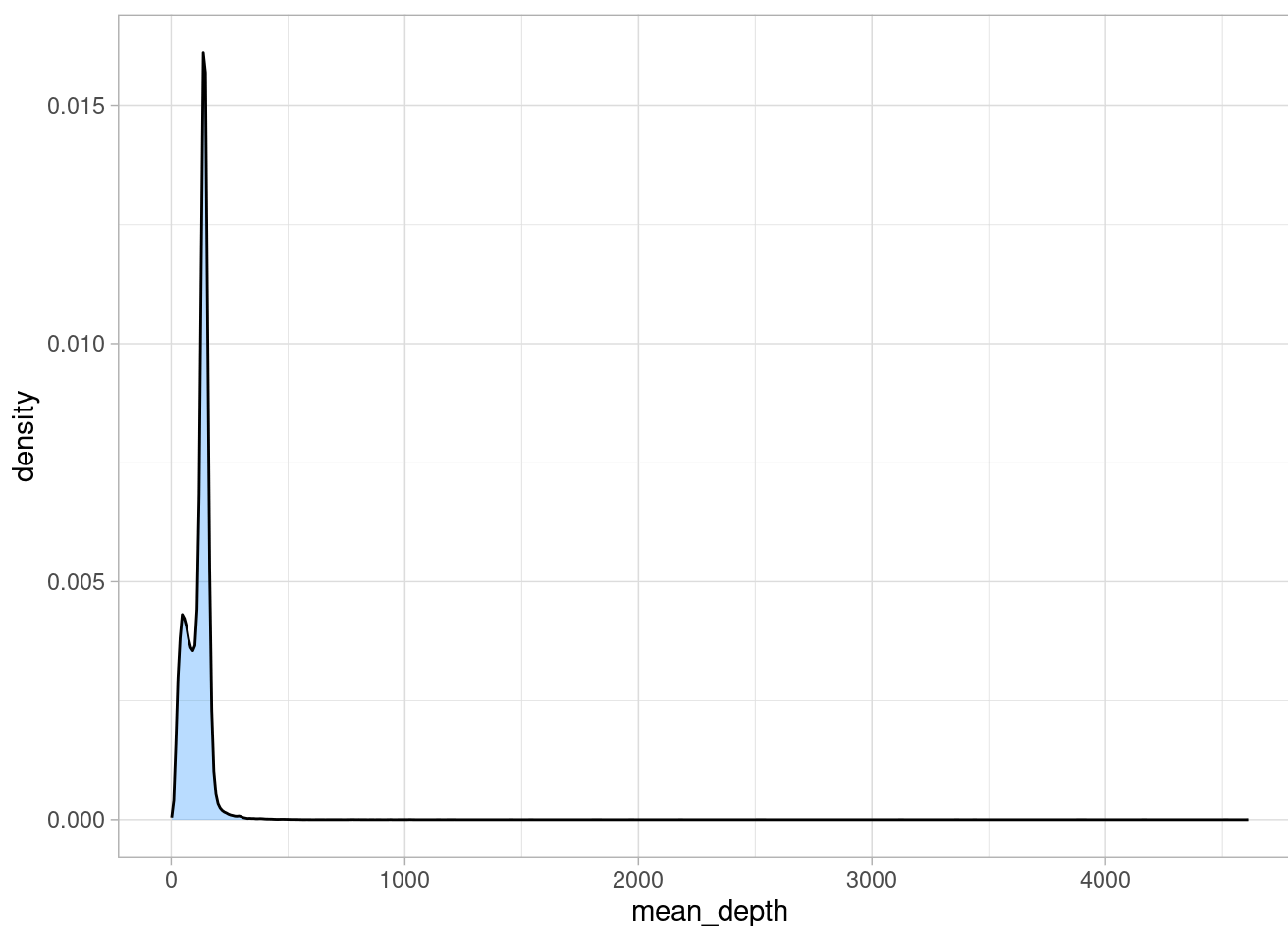
Variant Mean Depth

Mean number of reads mapped to each position.

```
var_depth <- read_delim("uflo_vcf_subset.ldepth.mean", delim = "\t",  
                        col_names = c("chr", "pos", "mean_depth", "var_depth"), skip =  
1)
```

```
## Rows: 59539 Columns: 4  
## — Column specification —————  
## Delimiter: "\t"  
## chr (1): chr  
## dbl (3): pos, mean_depth, var_depth  
##  
## i Use `spec()` to retrieve the full column specification for this data.  
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
a <- ggplot(var_depth, aes(mean_depth)) +  
  geom_density(fill = "dodgerblue1", colour = "black", alpha = 0.3)  
a + theme_light()
```

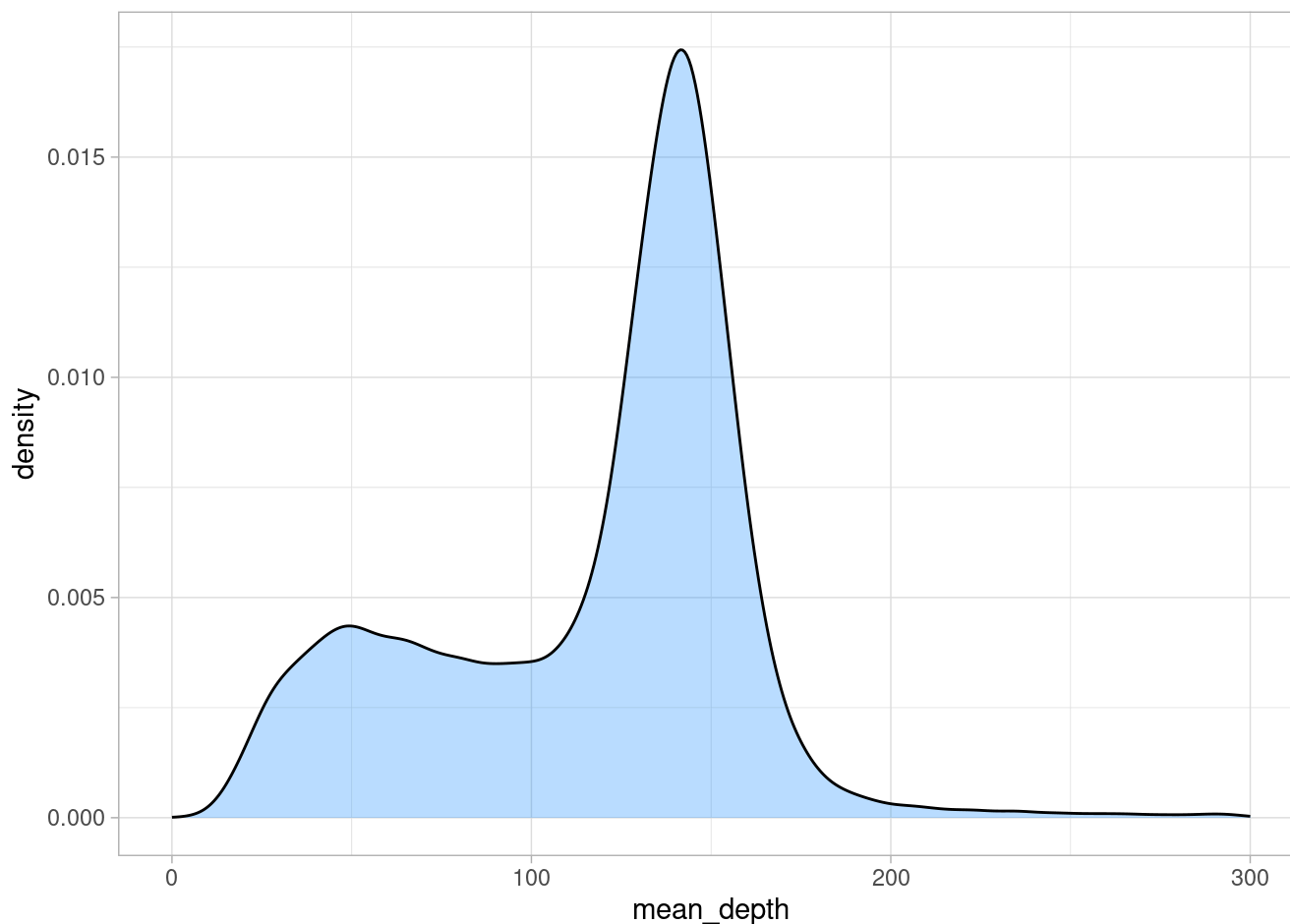


```
summary(var_depth$mean_depth)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      2.00   85.74  131.80  119.17  146.43 4610.97
```

```
a + theme_light() + xlim(0, 300)
```

```
## Warning: Removed 267 rows containing non-finite outside the scale range
## (`stat_density()`).
```



Decision: set min depth to 10, max depth to $\sim 2 \times$ mean ($119 * 2 \approx 240$). This handles outliers that could reflect paralogous or repetitive regions.

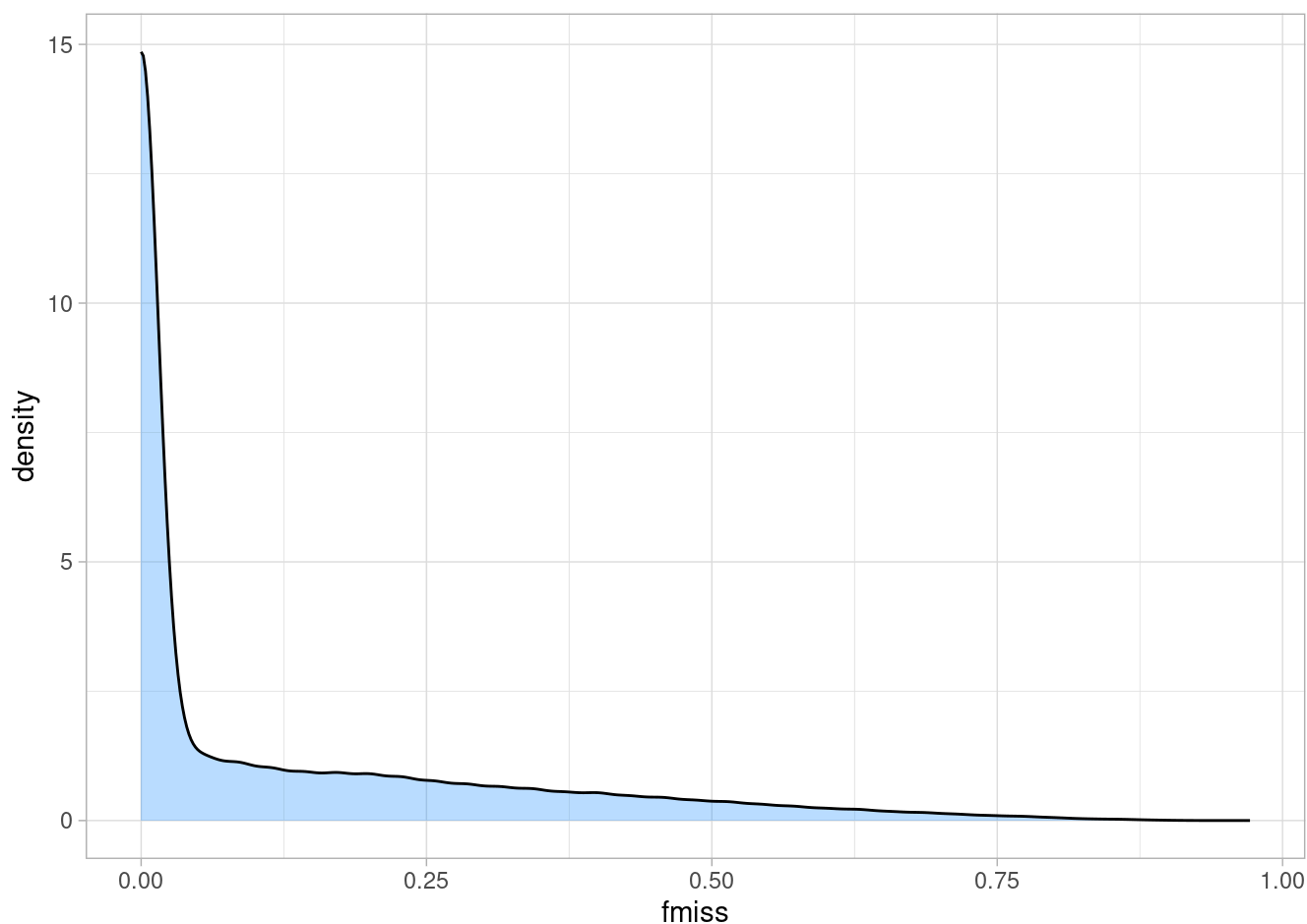
Variant Missingness

Proportion of individuals lacking a genotype at each call site.

```
var_miss <- read_delim("uflo_vcf_subset.lmiss", delim = "\t",
                      col_names = c("chr", "pos", "nchr", "nfiltered", "nmiss", "fmiss"), skip = 1)
```

```
## Rows: 59539 Columns: 6
## — Column specification —————
## Delimiter: "\t"
## chr (1): chr
## dbl (5): pos, nchr, nfiltered, nmiss, fmiss
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
a <- ggplot(var_miss, aes(fmiss)) +
  geom_density(fill = "dodgerblue1", colour = "black", alpha = 0.3)
a + theme_light()
```



```
summary(var_miss$fmiss)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.0000  0.0000   0.0000  0.1188  0.2000  0.9714
```

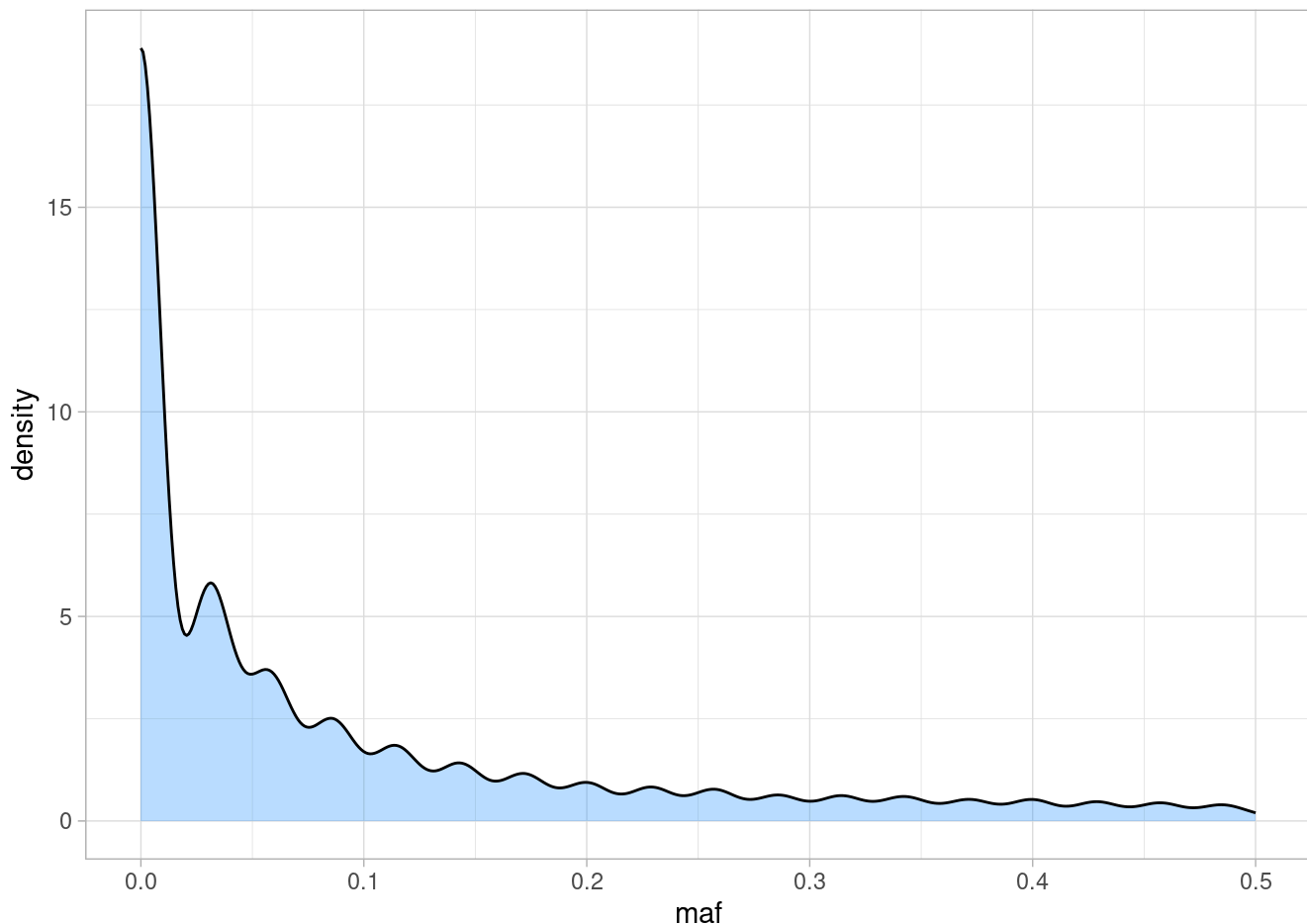
Decision: Typically 75–95% completeness is used as a threshold. Going with 90% (MISS = 0.10).

Minor Allele Frequency

```
var_freq <- read_delim("uflo_vcf_subset.frq", delim = "\t",  
                      col_names = c("chr", "pos", "nalleles", "nchr", "a1", "a2"), ski  
p = 1)
```

```
## Rows: 45756 Columns: 6  
## — Column specification —————  
## Delimiter: "\t"  
## chr (1): chr  
## dbl (5): pos, nalleles, nchr, a1, a2  
##  
## i Use `spec()` to retrieve the full column specification for this data.  
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
var_freq$maf <- var_freq %>%  
  select(a1, a2) %>%  
  apply(1, function(z) min(z))  
a <- ggplot(var_freq, aes(maf)) +  
  geom_density(fill = "dodgerblue1", colour = "black", alpha = 0.3)  
a + theme_light()
```



```
summary(var_freq$maf)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.00000 0.02941 0.08546 0.11429 0.50000
```

Decision: A large number of variants have low-frequency alleles. Will set MAF threshold to 0.1.

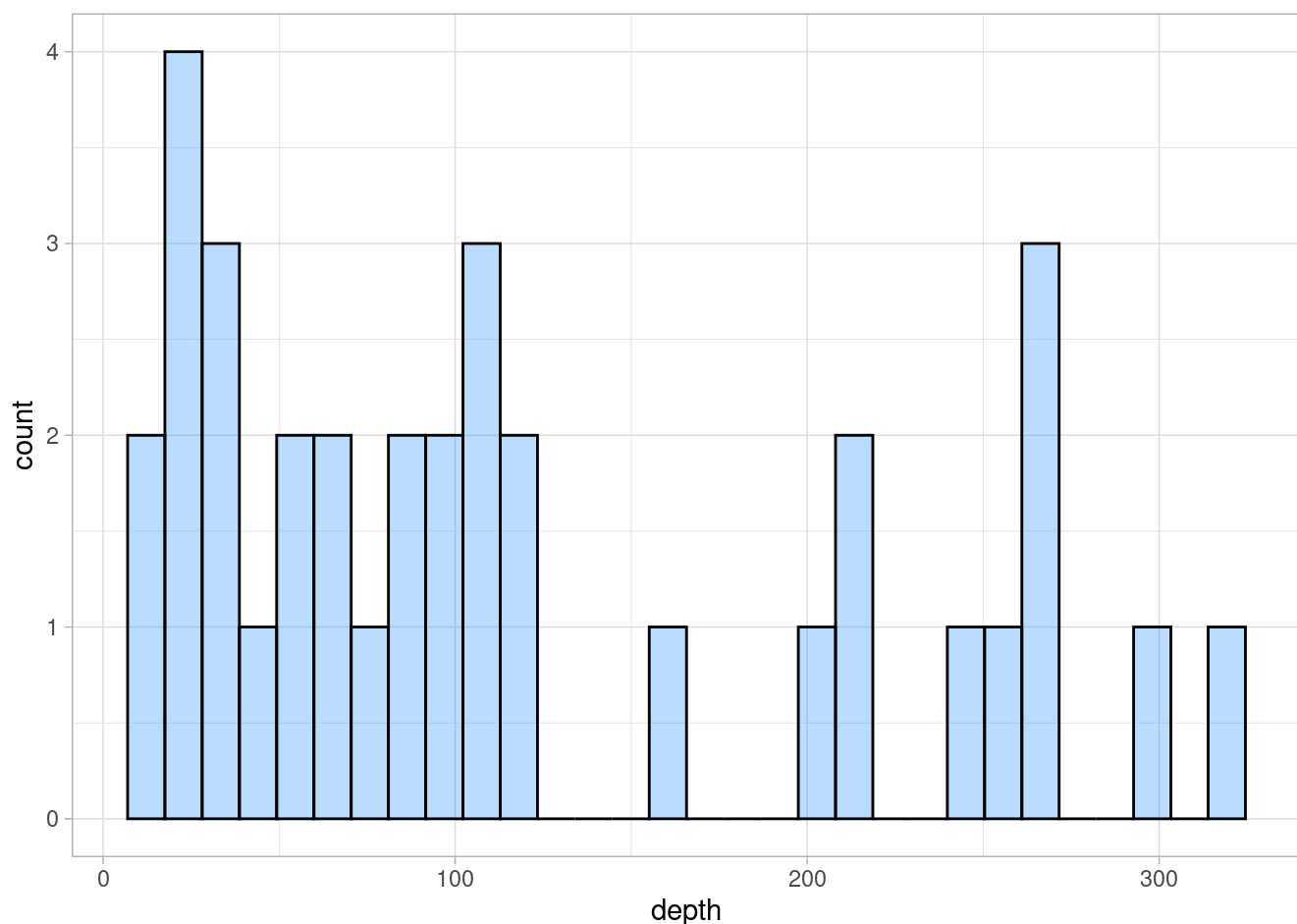
Mean Depth Per Individual

```
ind_depth <- read_delim("uflo_vcf_subset.iddepth", delim = "\t",
                        col_names = c("ind", "nsites", "depth"), skip = 1)
```

```
## Rows: 35 Columns: 3
## — Column specification —————
## Delimiter: "\t"
## chr (1): ind
## dbl (2): nsites, depth
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
a <- ggplot(ind_depth, aes(depth)) +
  geom_histogram(fill = "dodgerblue1", colour = "black", alpha = 0.3)
a + theme_light()
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



```
summary(ind_depth$depth)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      12.25  39.02   94.98  119.98 206.40  319.22
```

To note: Sequencing depth ranges from 12 to 319 — a bit of spread there.

Proportion of Missing Data Per Individual

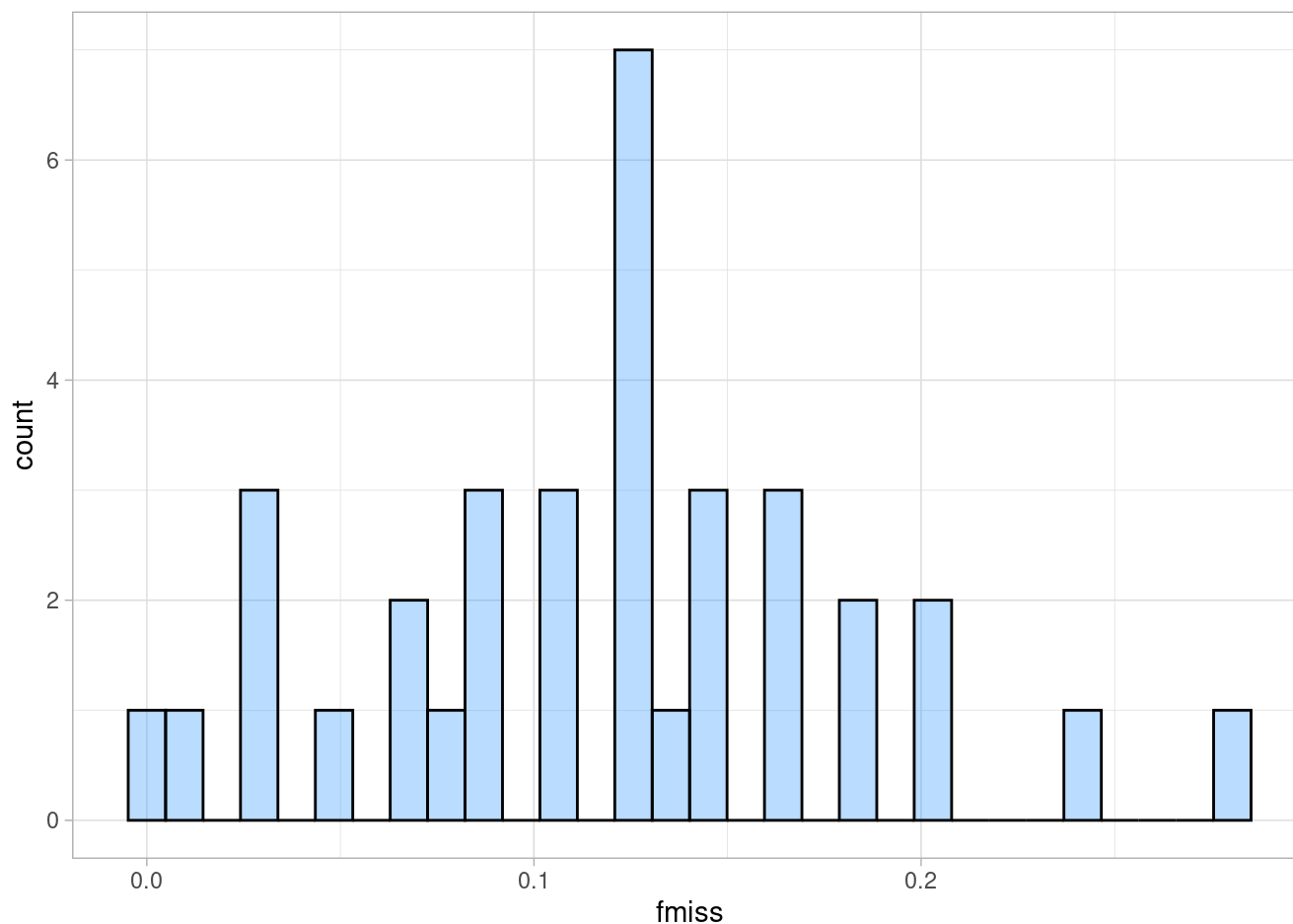
```
ind_miss <- read_delim("uflo_vcf_subset.imiss", delim = "\t",
                      col_names = c("ind", "ndata", "nfiltered", "nmiss", "fmiss"), skip
ip = 1)
```

```
## Rows: 35 Columns: 5
## — Column specification —————
## Delimiter: "\t"
## chr (1): ind
## dbl (4): ndata, nfiltered, nmiss, fmiss
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```



```
a <- ggplot(ind_miss, aes(fmiss)) +  
  geom_histogram(fill = "dodgerblue1", colour = "black", alpha = 0.3)  
a + theme_light()
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



```
summary(ind_miss$fmiss)
```

```
##      Min.   1st Qu.   Median     Mean   3rd Qu.     Max.   
## 3.359e-05 7.988e-02 1.232e-01 1.188e-01 1.541e-01 2.804e-01
```

To note: Missingness per individual ranges from 0 to 0.28.